Instructor Performance and Course Quality Evaluation on EduPro

Internship Project Report

Unified Mentor

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October 2026

1. Introduction

In online education, instructor quality and course design are among the strongest drivers of learner satisfaction, course ratings, repeat enrollments and platform credibility. Even when learner demand is strong, weak instruction can lead to low ratings, negative reviews and reduced trust in a platform.

EduPro is an online learning platform that needs a data-driven way to judge instructors and courses instead of relying on subjective impressions. This project analyses the EduPro dataset (teachers, courses and transactions) and delivers an interactive Streamlit dashboard that supports instructor evaluation, course quality monitoring and expertise-level insights.

2. Problem Statement

EduPro currently lacks clarity on four questions:

- Which instructors consistently deliver high-quality courses?
- Does teaching experience translate into better-rated courses?
- Are some course categories more dependent on instructor quality?
- How evenly is teaching performance distributed across the platform?

3. Objectives

- Integrate the Teachers, Courses and Transactions data into one analysis-ready dataset.
- Profile instructors by age, experience, expertise and rating, and identify top and low performers.
- Measure the relationship between experience and both teacher and course ratings.
- Evaluate course quality by category and level, including a gender comparison.
- Compare course ratings and enrollment volume across high, mid and low rated instructors.
- Identify expertise areas where teaching quality is strongest or most in need of training.
- Build a Streamlit dashboard with filters, a leaderboard, scatter plots and heatmaps.

4. Dataset Description

The data comes from one Excel workbook, *EduPro_Online_Platform.xlsx*, with four sheets. Three of them are used in this analysis.

Sheet	Rows	Key fields used
Teachers	60	TeacherID, TeacherName, Age, Gender, Expertise, YearsOfExperience, TeacherRating
Courses	60	CourseID, CourseName, CourseCategory, CourseLevel, CourseRating
Transactions	10,000	TransactionID, CourseID, TeacherID
Users (not used)	3,000	UserID, UserName, Age, Gender, Email

There are no missing values in the three sheets used. The data has 12 expertise areas, 12 course categories and 3 course levels (Beginner, Intermediate, Advanced).

5. Methodology

Tools: Python, pandas for data handling, Plotly for dashboard charts, and Streamlit for the dashboard, developed in Visual Studio Code.

Step 1 - Data integration. The Courses sheet does not contain a TeacherID. The only link between instructors and courses is the Transactions sheet, which holds both CourseID and TeacherID. Transactions was joined to Teachers on TeacherID and to Courses on CourseID, giving 10,000 merged rows with no unmatched records. Because a course can be taught by several instructors in the transaction data, enrollments are counted per teacher and course pair.

Step 2 - Instructor profiling. Distributions of age, experience, expertise and rating were computed, and instructors were ranked by TeacherRating.

Step 3 - Experience versus performance. Pearson correlations were calculated between YearsOfExperience and TeacherRating, between YearsOfExperience and each instructor's average CourseRating, and between TeacherRating and average CourseRating. Instructors were also grouped into experience bands to look for thresholds.

Step 4 - Course quality. CourseRating was analysed by category, by level, and by instructor gender and level, and shown as a category by level heatmap.

Step 5 - Instructor tiers. Instructors were grouped by TeacherRating into High (4.0 and above), Mid (3.0 to 3.99) and Low (below 3.0). These cut-offs were chosen by the analyst for clarity.

Step 6 - Expertise insights. Teacher rating, course rating and enrollments were compared by expertise area.

6. Key Performance Indicators

KPI	Definition used	Value
Average Teacher Rating	Mean TeacherRating across instructors	3.12
Average Course Rating	Mean CourseRating across courses	3.10
Rating Consistency Index	Average of 1 minus (std. deviation of an instructor's course ratings / 5)	0.77
Experience Impact Score	Correlation between years of experience and teacher rating	0.60
Enrollment Influence Ratio	Share of enrollments going to High-tier instructors divided by their share of instructors	3.72

The Rating Consistency Index runs from 0 to 1, where 1 means all of an instructor's courses are rated identically. An Enrollment Influence Ratio above 1 means highly rated instructors attract more than their proportional share of demand.

7. Findings and Analysis

7.1 Instructor profile and rating distribution

The platform has 60 instructors (33 male, 27 female) aged 27 to 50, with an average age of about 38. Experience ranges from 1 to 24 years with a median of 6 years. Teacher ratings average 3.12 (median 3.28, standard deviation 0.95) and range from 1.05 to 4.97.

Rating band	Instructors
2.0 or below	9
Above 2.0 to 3.0	12
Above 3.0 to 4.0	28
Above 4.0	11

Most instructors sit in the 3 to 4 range, but 21 of 60 (35 percent) are rated 3.0 or below, so quality is not evenly distributed. Instructors are also concentrated in a few areas: Digital Marketing has 11 and Cybersecurity has 9, while Marketing and Programming have only 1 each. The highest rated instructor is Yolanda Levine (Machine Learning, 4.97), followed by Kimberly Miller (Cybersecurity, 4.58). The lowest rated is Timothy Bailey (Business, 1.06).

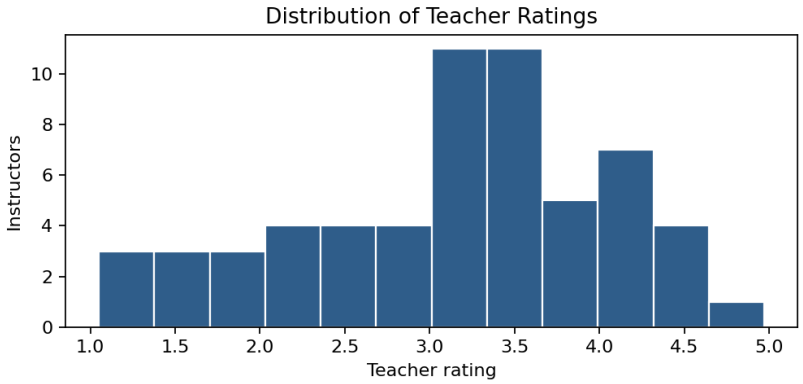


Figure 1: Distribution of teacher ratings

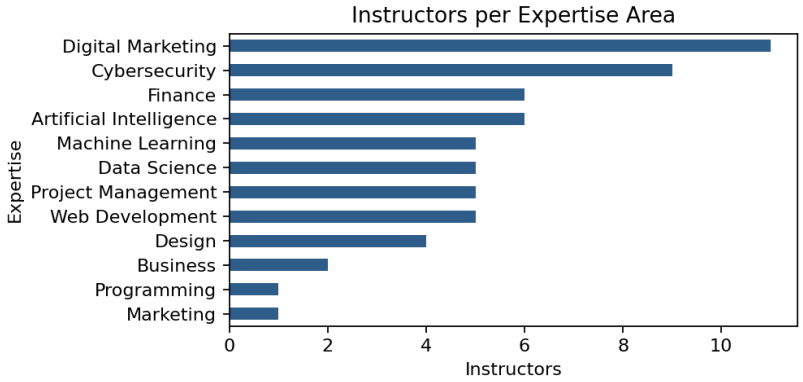


Figure 2: Instructors per expertise area

7.2 Experience versus performance

- Experience and teacher rating show a moderate positive correlation of **0.60** (Spearman 0.67): more experienced instructors tend to be rated higher.
- Experience and the average course rating of an instructor's courses show **no relationship** (correlation -0.06).
- Teacher rating and average course rating are also **unrelated** (correlation 0.00).

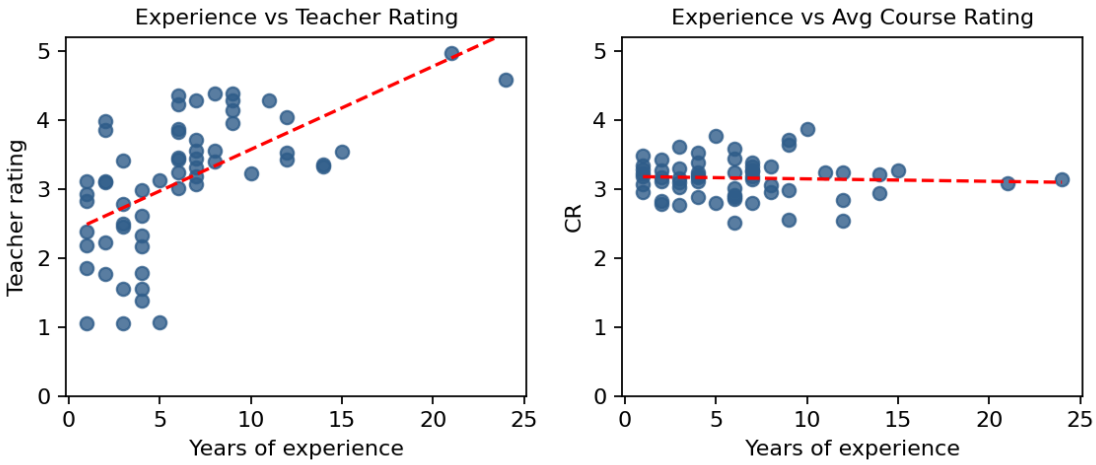


Figure 3: Experience versus teacher rating and average course rating

Experience band	Instructors	Avg teacher rating	Avg course rating
0-2 years	13	2.64	3.16
3-5 years	15	2.19	3.20
6-8 years	18	3.63	3.12
9-12 years	9	3.92	3.18
13+ years	5	3.95	3.13

Teacher ratings jump sharply at about 6 years of experience and flatten after about 9 years, which suggests an experience threshold followed by diminishing returns. Course ratings stay flat at about 3.1 to 3.2 in every band.

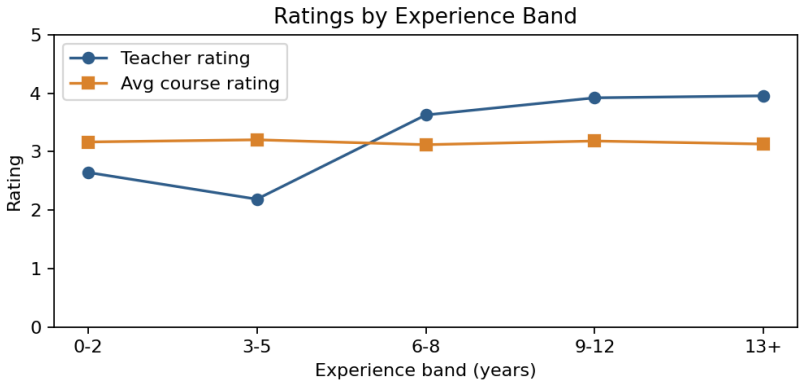


Figure 4: Ratings by experience band

7.3 Course quality

Course ratings average 3.10 with a wide spread (standard deviation 1.17, range 1.13 to 4.94).

By category. Marketing (3.72) and Digital Marketing (3.66) are the highest rated categories. Machine Learning (2.66), Business (2.69) and Web Development (2.85) are the lowest. Data Science (std. deviation 0.50) and Business (0.73) are the most consistent categories, while Project Management (1.71) and Machine Learning (1.56) vary the most.

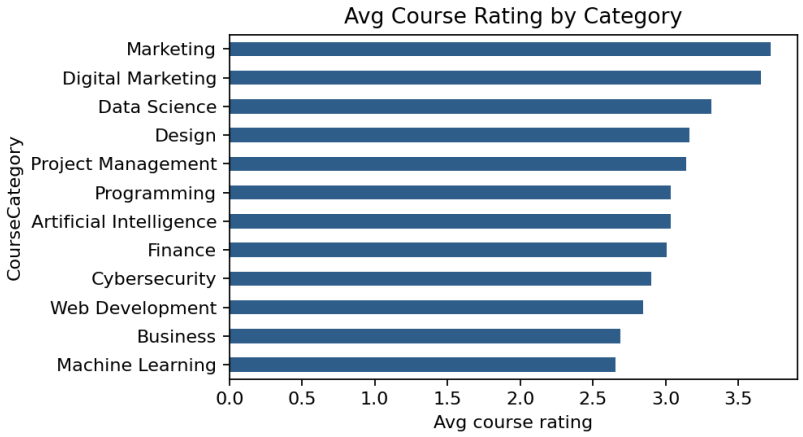


Figure 5: Average course rating by category

By level. Intermediate courses rate highest (3.32), followed by Beginner (3.19). Advanced courses rate lowest (2.81), which suggests advanced content needs the most quality attention.

Category by level heatmap. Ratings vary widely inside a single category. Cybersecurity Beginner courses average 4.94 while its Advanced and Intermediate courses average about 2.4, and Project Management Intermediate averages 4.14 against 1.23 for Advanced. This points to course-level design differences rather than category-wide quality.

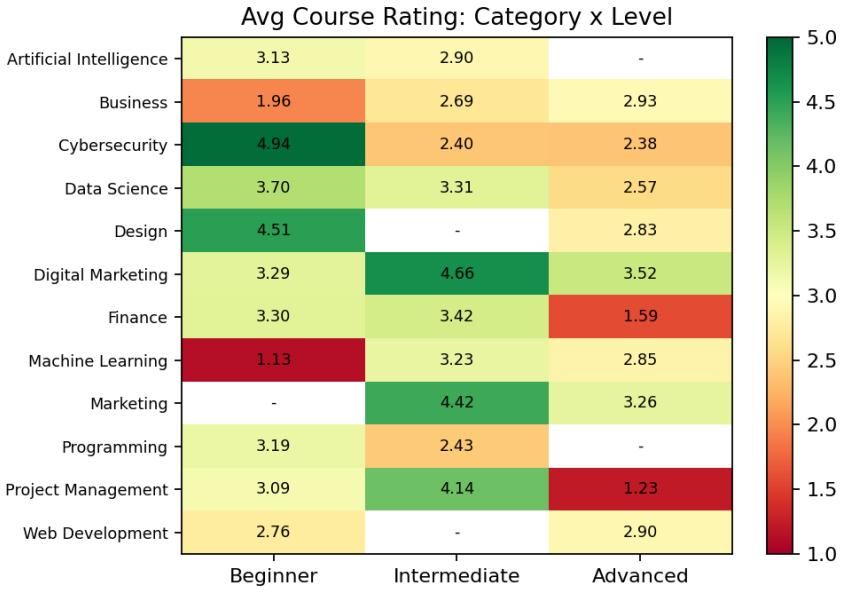


Figure 6: Category by level heatmap of average course rating

Gender and level. Courses taught by male instructors average 3.19 and by female instructors 3.06. The gap is mainly at Beginner level (3.37 male against 3.10 female). At Intermediate (3.34 and 3.33) and Advanced (2.84 and 2.80) levels the two groups are almost identical.

7.4 Instructor impact on course success

Instructor tier	Instructors	Avg course rating	Total enrollments	Enrollments per instructor
Low (below 3.0)	21	3.14	1,604	76
Mid (3.0 to 3.99)	28	3.08	1,576	56
High (4.0 and above)	11	3.13	6,820	620

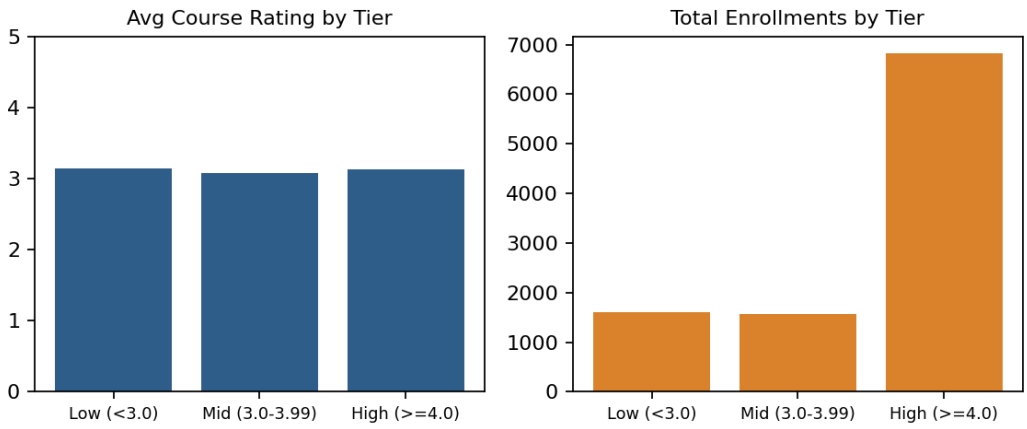


Figure 7: Course rating and enrollments by instructor tier

Course ratings do not differ by instructor tier. High, mid and low rated instructors all teach courses rated about 3.1, which agrees with the zero correlation between teacher and course rating.

High rated instructors attract far more enrollments, but this is driven by two people. Yolanda Levine (3,061 enrollments) and Kimberly Miller (3,025) together account for 6,086 of the 10,000 enrollments (about 61 percent). The other nine High-tier instructors average only about 82 enrollments each, similar to the other tiers. Across all instructors the correlation between teacher rating and enrollments is a modest 0.32. The Enrollment Influence Ratio of 3.72 should therefore be read with care, since it reflects heavy concentration of demand rather than a broad effect of high ratings.

7.5 Expertise-based insights

Expertise	Instructors	Avg teacher rating	Avg course rating	Enrollments
Marketing	1	4.29	3.65	120
Machine Learning	5	4.23	2.78	3,303
Programming	1	3.95	2.98	123
Cybersecurity	9	3.47	3.04	3,450
Finance	6	3.26	3.17	410
Artificial Intelligence	6	3.17	3.16	389
Digital Marketing	11	3.05	3.55	479
Data Science	5	3.03	3.28	432
Project Management	5	2.66	3.08	368
Web Development	5	2.51	2.94	344
Design	4	2.29	3.16	324
Business	2	2.24	2.80	258

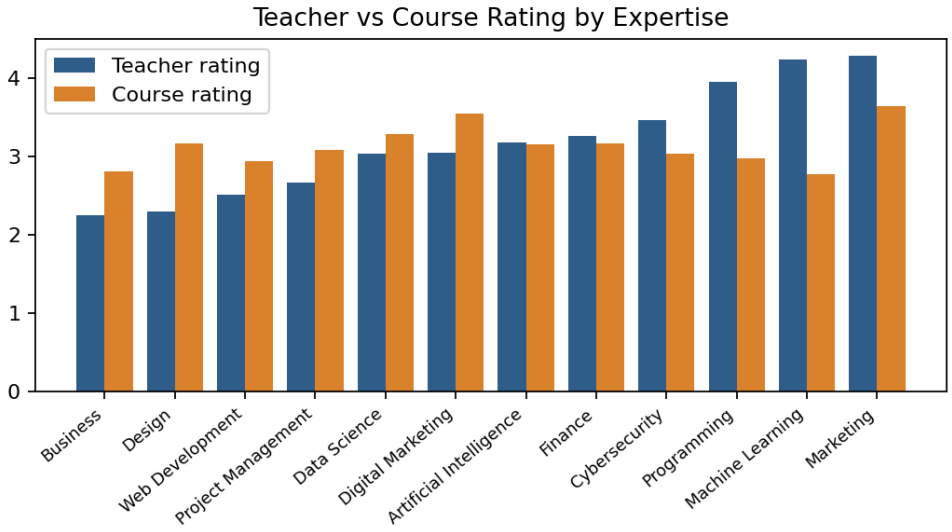


Figure 8: Teacher rating versus course rating by expertise

- **Strongest instructors:** Machine Learning (4.23) and Cybersecurity (3.47), which also carry most of the platform's enrollments (6,753 combined). Marketing and Programming score well but have only one instructor each, so they are not reliable conclusions.
- **Biggest mismatch:** Machine Learning has the highest rated instructors but the second-lowest course rating (2.78), so great instructors are not producing equally well-received courses there. Digital Marketing shows the opposite: modest instructor ratings (3.05) but strong course ratings (3.55).
- **Training needs:** Business, Design, Web Development and Project Management have the lowest average teacher ratings (all below 2.7) and are the clearest candidates for instructor development.
- **Where instructor quality matters most:** the link between teacher rating and course rating inside an expertise area varies a lot, but each area has only a handful of instructors, so these patterns are indicative only.

8. Dashboard Overview

The Streamlit application (app.py) reads the Excel workbook, performs the data integration and presents the results in seven tabs:

- **Leaderboard:** top and low performing instructors with rating, course rating, enrollments and consistency.
- **Instructor Profile:** histograms of rating, experience and age, expertise counts and rating spread.
- **Experience vs Rating:** scatter plots with trend lines, correlation values and the experience band chart.
- **Course Quality:** category by level heatmap, category and level comparisons, and gender versus level.
- **Instructor Impact:** rating tier comparison of course ratings and enrollments.
- **Expertise Insights:** expertise-wise comparisons and training needs.
- **Data Check:** validation of the table joins.

Users can filter the whole dashboard by instructor expertise, course category, course level, teacher rating range and course rating range. The five KPIs are shown at the top and update with the filters.

[Insert your dashboard screenshots here: home page with KPI row and sidebar filters, Leaderboard tab, and any other tabs you want to show.]

9. Recommendations

1. **Do not rely on experience alone to judge course quality.** Experience improves learner ratings of the instructor up to about 6 to 9 years, but has no visible effect on course ratings. Course quality needs its own review process.
2. **Invest in training for low-rated expertise areas.** Business, Design, Web Development and Project Management should be prioritised for coaching and mentoring by top performers.
3. **Review Advanced courses.** They have the lowest average rating (2.81), with some very low scoring combinations such as Project Management and Finance Advanced courses.
4. **Investigate the Machine Learning gap.** Highly rated instructors are delivering poorly rated courses, which suggests a content design or difficulty problem rather than a teaching problem.
5. **Reduce demand concentration.** About 61 percent of enrollments go to two instructors. Promoting other strong instructors would lower reliance on a few individuals.
6. **Track the KPIs regularly** through the dashboard to monitor changes in consistency and quality.

10. Limitations

- The Courses sheet has no direct instructor link, so the instructor to course mapping relies on the Transactions sheet, where one course can appear with several instructors.
- The dataset is small (60 instructors and 60 courses, with 1 to 5 courses per category), so correlations and category differences are indicative rather than statistically conclusive.
- Enrollments are heavily concentrated in two instructors, which strongly influences the enrollment comparisons.
- The tier cut-offs and the Rating Consistency Index formula are analyst-defined choices.
- Correlation does not prove cause; higher ratings may follow from popularity as much as the reverse.

11. Conclusion

The analysis shows that instructor evaluation on EduPro can be made objective and measurable. Teacher ratings rise with experience up to roughly 9 years, but neither experience nor teacher rating predicts how well a course is rated, so instructor quality and course quality must be monitored separately. Course quality differs more by individual course, level and category than by instructor tier, with Advanced content and the Machine Learning, Business and Web Development categories needing the most attention. Highly rated instructors attract more learners, but demand is concentrated in two instructors. The Streamlit dashboard gives EduPro a reusable tool to explore these insights with filters and to track the five KPIs over time.